dryness of the fabric that causes the static charge build up. Anti-static effective chemicals are largely chemically inert and require thermosol or heat treatment for fixing. In general thermsolable anti-static agents also have a good soil release action which is as permanent as the anti-static effect. Anti-static finishes may also be of polyamide type being curable at moderate temperatures.

1.3.18 Moth Proof Finish

Moth proofing finish is a kind of functional finish given to textiles to prevent the growth of moths. Moths like silverfish attack fibres like cotton and wool. Fluorine compounds, napthalene, DDT and paradichloro benzene are some of the chemicals used for imparting moth proof finishes to fabrics. They are available in crystal, cake and spray form. Cellulosic fibres are also treated with boric acid to prevent the rapid growth of mildew and fungus.



Figure 1.16 A Woollen Fabric Eaten by Moths

1.3.19 Shrink Proofing or Sanforising or Compacting

Controlled residual shrinkage is an important quality parameter for many fabrics. For example, excessive shrinkage is undesirable for fabrics to be made into garments. Here, the residual shrinkage should be less than 2% otherwise the garment will not fit after it is laundered. Mechanical compacting is one method of reducing residual shrinkage. The process forces yarns closer together and the fabric becomes thicker and heavier. As a result of this, the net yardage yield is reduced. A sanforizer is a fabric compactor developed by Cluett Peabody. The term 'Sanforized", is their registered trademark and is used to market fabrics that meet certain shrinkage specifications. The term Sanforized is now generally accepted to mean a fabric that has low residual shrinkage. It is used to describe shrink proofing processes. The process, consists of a range where the fabric is first moistened with steam, to make it more pliable, run through a short tenter frame to straighten and smooth out wrinkles ,through the compressive shrinkage head and then through a Palmer drying unit to set the fabric.

The key to any compactor is the head where force is applied to move parallel yarns closer together. More fabric must be fed in than is taken off. A Sanforizer uses a thick rubber blanket running against a steam heated cylinder as the compacting force. The thick rubber blanket first goes over a smaller diameter roll which stretches the convex surface of the blanket. Fabric is metered onto the stretched blanket and the fabric and blanket together come in contact with the steam heated cylinder. At this point, the stretched rubber surface contracts to its original length and then is forced to contract an additional amount as it forms the concave configuration of the heated drum. Since the fabric is not elastic, an extra length of fabric is thrust between the rubber

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